



RV College of
Engineering®

Experiential Learning Phase 1

Department of Computer Science,
R V College of Engineering, Bengaluru.

Dual-Mode Secure API Communication with Multi- Metric Driven Protocol Selection

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What is the Problem?

Modern communication systems rely on fixed protocols such as TLS 1.3, which introduce handshake and encryption overhead that can impact performance under varying network conditions. While optimized protocols like QUIC improve efficiency, they still operate within a single protocol framework without adapting across different communication strategies. Existing systems do not dynamically balance security and performance based on real-time network behavior, leading to suboptimal resource utilization.

What is the Solution?

An Adaptive Session-Based Protocol Selection System that dynamically chooses between:

- Secure Mode — TLS-based communication ensuring strong security guarantees
- Fast Mode — a custom lightweight TCP protocol with reduced overhead

The system uses a multi-metric decision engine (latency, handshake cost, payload size) to select the optimal protocol at the session level. It also incorporates a hysteresis-based switching mechanism to ensure stable and efficient adaptation without frequent protocol changes.

Literature Review

SR NO.	TITLE	AUTHORS	YEAR	INFERENCE
1.	Performance Evaluation of SSL/TLS Handshake Latency in Distributed Web Service Architectures	T M Sathish Kumar	2025	• This paper analyzes SSL/TLS handshake latency in distributed web service systems using modern transport protocols such as HTTP/2 and QUIC. • Experimental results across 100 globally distributed cloud servers show that TLS 1.3 reduces handshake latency by 35% compared to TLS 1.2, due to fewer round trips and a streamlined handshake process. • The study discusses the trade-off between security robustness and latency efficiency in distributed systems.
2.	Evaluating Transport Layer Security 1.3 Optimization Strategies for 5G Cross-Border Roaming: A Comprehensive Security and Performance Analysis	Jhury Kevin Yongho Ko Hoseok Kwon Ilsun You	September 2025	• This paper analyzes the performance and security trade-offs of TLS 1.3 in 5G cross-border roaming. • It shows that existing TLS 1.3 techniques either reduce latency at the cost of security or maintain strong security with increased handshake latency. • The authors propose 0-RTT Forward Secrecy (0-RTT FS), enabling zero round-trip handshakes while maintaining forward secrecy and replay protection. • The proposed method achieves very low handshake latency (~195 μs) with minimal overhead and resolves vulnerabilities in existing TLS optimization techniques.

SR NO.	TITLE	AUTHORS	YEAR OF PUBLICATION	INFERENCE
3.	Comparative analysis of post-quantum handshake performance in QUIC and TLS protocols	José A. Montenegro Ruben Rios Javier Bonilla	2026	- Compares TLS and QUIC with post-quantum cryptography, analyzing performance under different network conditions. - Hybrid KEMs increase latency and overhead, while pure post-quantum KEMs provide a better security- performance balance. - QUIC improves handshake efficiency and reduces delays compared to TLS, especially in lossy networks. - Highlights the trade-offs and feasibility of adopting quantum-resistant communication protocols.
4.	Performance and Security Assessment of Transport Layer Security (TLS) Protocol Versions	Aqib Javed	October 2024	- Compares TLS versions (1.0-1.3) in terms of performance and security, highlighting the trade-off between them. - Shows TLS 1.3 significantly reduces handshake latency and improves overall performance. - Identifies vulnerabilities in older TLS versions and highlights stronger security features in TLS 1.3. - Provides guidance on selecting appropriate TLS versions based on performance and security needs.

SR NO.	TITLE	AUTHORS	YEAR OF PUBLICATION	INFERENCE
5.	Analysis of Next-Generation Internet Transport Protocols: QUIC, WebTransport, HTTP/3	O.M. Bondar	December 2025	- Analyzes next-generation transport protocols—QUIC, HTTP/3, and WebTransport—designed to overcome TCP/HTTP limitations like head-of-line blocking and multi-RTT handshakes. - Highlights QUIC's 1-RTT handshake, encrypted headers, and improved congestion control for low-latency communication. - Shows WebTransport extends HTTP/3 with multiplexed streams and unreliable datagrams, supporting modern applications like cloud gaming and streaming. - Notes practical deployment benefits and QUIC libraries, while open issues remain in multicast, satellite links, and observability.
6.	A Quantum of QUIC: Dissecting Cryptography with Post-Quantum Insights	Marcel Kempf Nikolas Gauder Benedikt Jaeger Johannes Zirngibl	February 2024	- Evaluates QUIC's performance and security under different cryptographic schemes, including post-quantum algorithms. - Shows removing packet protection increases performance by 10-20%, while header protection has minimal impact, especially with AES. - Integrates post-quantum algorithms into QUIC, demonstrating feasibility with low handshake overhead. - Notes heavy algorithms that significantly increase handshake duration and reduce performance.

SR NO.	TITLE	AUTHORS	YEAR OF PUBLICATION	INFERENCE
7.	Challenges and Advances in Analyzing TLS 1.3-Encrypted Traffic: A Comprehensive Survey	Jiuxing Zhou Wei Fu Wei Hu Zhihong Sun Tao He Zhihong Zhang	October 2024	- Reviews challenges in analyzing TLS 1.3 encrypted traffic due to enhanced security features. - Shows traditional traffic analysis methods (used for TLS 1.2) are ineffective, requiring new approaches. - Explores modern techniques such as middlebox-based methods, searchable encryption, and machine learning based analysis. - Highlights the trade-off between strong privacy/security and the need for effective traffic monitoring and anomaly detection.
8.	Revisiting QUIC attacks: a comprehensive review on QUIC security and a hands-on study	Efstratios Chatzoglou Vasileios Kouliaridis Georgios Karopoulos Georgios Kambourakis	June 2025	- Reviews QUIC's security features and compares it with the traditional TCP + TLS + HTTP/2 model. - Highlights benefits like reduced connection time, multiplexing, and always-encrypted communication. - Identifies practical vulnerabilities through fuzz testing, including zero-day attacks that can overwhelm servers. - Concludes that QUIC implementations are still maturing, with security risks despite strong design advantages.

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9.	Delay Measurement of 0-RTT Transport Layer Security (TLS) Handshake Protocol	Danylo Goncharsky Sung Yong Kim Ahmed Serhrouchni Pengwenlong Gu Rida Khatoun Joel Hachem	2022	0-RTT reduces connection setup time drastically — up to 83x faster than legacy 2-RTT. This means the biggest latency cost in TLS is not encryption itself, but the round-trip negotiation. A custom protocol that skips multi-round handshakes can achieve similar speedups without needing post-quantum cryptography.
10.	The impact of data-heavy, post-quantum TLS 1.3 on the Time-To-Last-Byte of real-world connections	Panos Kampanakis Will Childs-Klein	2024	- TLS handshake cost is not constant — it grows non-linearly as RTT and packet loss increase. A large certificate chain that adds one extra round-trip at 35ms RTT becomes significantly worse at 70ms RTT - The paper uses netem (Linux kernel traffic control) for network emulation, which requires root access and a Linux environment. This project uses Python-level delay injection instead, which is less precise and doesn't emulate packet loss or bandwidth caps.



1. **Lack of Cross-Protocol Adaptive Selection Mechanisms**
Existing works focus on optimizing individual protocols such as TLS 1.3 and QUIC, but do not explore dynamic switching between different protocol types based on real-time network conditions [5], [8].
2. **Focus on Handshake Optimization Rather Than Adaptive Decision Systems**
Several studies analyze and reduce handshake latency (e.g., TLS performance and o-RTT optimization), but they do not utilize these metrics to drive runtime protocol selection [1], [2], [9].
3. **Absence of Multi-Metric Decision Models for Protocol Selection**
Most research evaluates performance using isolated parameters such as latency or security, without proposing a combined multi-metric decision framework for adaptive communication [4], [7].
4. **Limited Work on Stability-Aware Adaptive Switching**
Existing literature does not address the issue of instability caused by frequent switching, and lacks mechanisms like hysteresis-based control for stable protocol adaptation [3], [6].



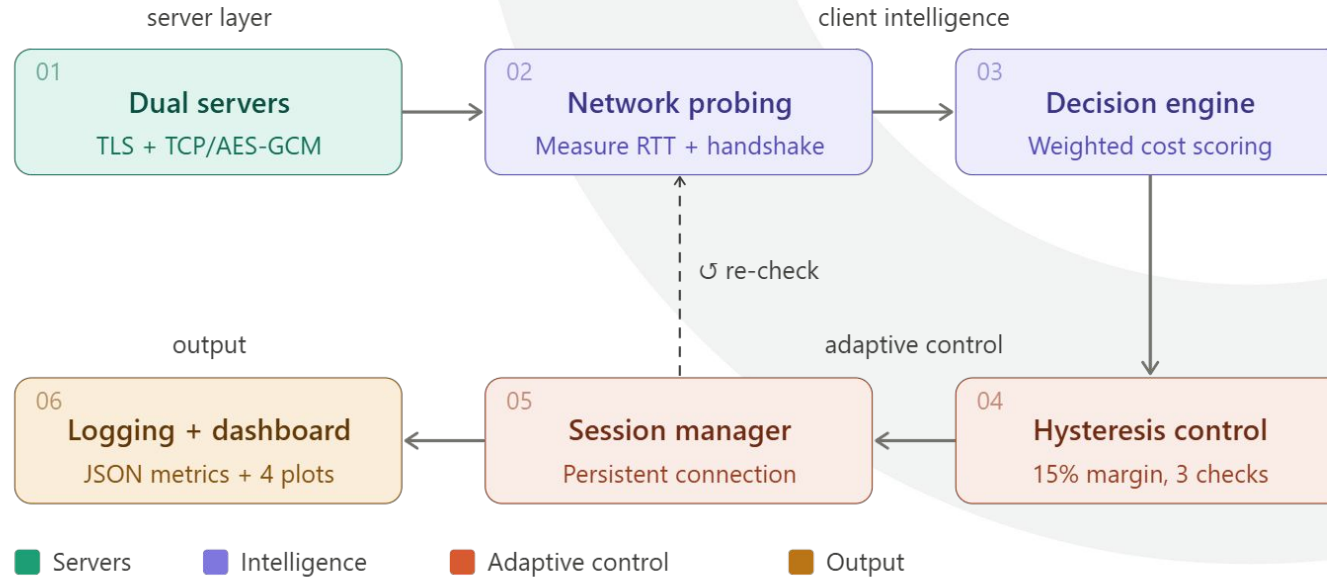
Problem Statement

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Modern API communication systems use fixed protocols that do not adapt to changing network conditions, leading to inefficiencies in performance and security trade-offs. High latency and handshake overhead can significantly impact response time in such systems. There is a need for an adaptive approach that can intelligently choose between secure and lightweight communication methods. This project aims to address this by designing a system that dynamically optimizes protocol selection based on real-time network behavior.



- Design a client–server system supporting both TLS and custom TCP protocols.
- Implement a multi-metric decision engine to select the optimal protocol.
- Ensure stable switching between protocols using a hysteresis mechanism.
- Collect and analyze performance and security metrics for comparison.





1. Dual Server Implementation

- TLS Server: Flask over HTTPS (port 5000), TLS 1.3, JSON API
- TCP Server: Raw socket (port 6000), AES-GCM encrypted payloads, custom HELLO»ACK handshake
- Both expose identical logical API — only transport differs

2. Network Probing

- Client probes both protocols before session starts
- Measures: RTT (round-trip time) and handshake duration for each
- Artificial delay injected to simulate low (20ms) / high (180ms) latency

3. Decision Engine (Scoring Model)

- Computes cost score for each protocol:
$$\text{Score} = w_1 \times \text{latency} + w_2 \times \text{handshake_time} + w_3 \times \text{payload_cost}$$
- Lower score = preferred protocol



4. Hysteresis Controller
 - Candidate protocol must show >15% score advantage
 - Must hold that advantage for 3 consecutive checks
 - Switching only at safe points (after request completion)
 - Prevents oscillation under unstable network conditions
5. Session Manager
 - Maintains one active connection (HTTP keep-alive or persistent TCP socket)
 - Re-evaluates every 5 requests or 10 seconds
 - On switch: closes old connection cleanly, opens new one
6. Logging & Visualization
 - Every request logs: timestamp, protocol, RTT, handshake time, payload size, both scores
 - Dashboard plots: protocol timeline, RTT comparison, handshake overhead, score divergence



- Successful implementation of a dual-protocol client-server system supporting both TLS and a custom TCP-based communication model.
- Development of an adaptive decision engine capable of selecting the optimal protocol based on real-time network metrics.
- Demonstration of stable protocol switching using a hysteresis mechanism, avoiding frequent and unnecessary transitions.
- Collection and visualization of performance data to provide a comparative analysis of latency, handshake overhead, and protocol efficiency.
- Clear insights into the trade-offs between security and performance, enabling better understanding of adaptive communication strategies.



Conclusion

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- The project successfully developed an adaptive client-server API system that dynamically switches between a secure mode using TLS and a lightweight custom protocol using AES based on network latency (RTT).
- The system demonstrated that while TLS ensures strong security, it introduces higher latency, whereas the lightweight protocol reduces communication overhead and improves performance under high-latency conditions.
- The adaptive mechanism effectively selected the appropriate protocol at runtime, achieving a balance between security and performance across varying network conditions.
- The use of logging and visualization tools (e.g., Chart.js) enabled clear analysis of protocol behavior, latency trends, and switching patterns, providing meaningful insights into system performance.



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